MTH 231(C) Dr. James Ashe
Exam 2 November 9, 2012 **Name:**

Find the displacement over the given interval.

1. $v(t) = \cos\left(\frac{\pi}{2}t\right)$ on $[0, 3]$; $s(0) = 0$.
[height=6cm, width=6cm, axis lines=middle, xmin=-1,xmax=3, ymin=-1,ymax=10, xlabel=x,ylabel=y, area style,tension=0.08]
+[domain=0:2,fill,black!30!white]4*sqrt(2*x) ; +[domain=-1:3,fill,white]-4*x+6 ; +[domain=-1:3,fill,white]2*x^2;
[domain=0:3,thick,smooth,samples=500]4*sqrt(2*x) node[anchor=135] $f(x) = \frac{1}{x}$; [domain=-1:3,thick]
-4*x+6 node[right] $f(x) = \frac{2}{x}$; [domain=-1:3,thick] 2*x$^2 node[right]$f(x)=x;
at (axis cs:1,1.5) R;

Determine the position function.

2. An object is moving at a velocity of $(t + 9)^{1/2}$ with an initial position of $s(0) = 20$ at time t.

Solve the problem.

3. A projectile is launched vertically from the ground at $t = 0$ where t is seconds, and its velocity in flight (in m/s) is given by $v(t) = 20 - 10t$.

a. Find the position at time t for $0 \le t \le 4$.

b. Find the total distance traveled from $t = 0$ to $t = 4$.

Use calculus to find the area of the region described.

4. The region bounded by $y = x$ and $y = x^2 - 4x$.

D:/home/jim/JamesAshe/JCSU/Calc_2_MTH_232/images/x_and_x2-4x.png

5. The region bounded by $y = \frac{1}{2}x^2$ and $y = -x^2 + 6$.

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6. The region in the first quadrant bounded by $-x + 1$ and $x - 1$ from $0 \leq x \leq 2$.

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Use the general slicing method and calculus to find the volume of the solid.

7. The rectangular box with dimensions a height of 3, width of 4, and length of 10.

Let R be the region bounded by the following curves. Use the disk method to find the volume of the solid generated when R is revolved about the x-axis.

8. $y = \sqrt{x}$, $x = 0$, and $x = 9$.

9. $y = \sqrt{\sin 4x}$, $y = 0$, $0 \leq x \leq \dfrac{\pi}{4}$.

Find the length of the curve.

10. $y = 2x^{3/2}$ from $x = 0$ to $x = \dfrac{5}{4}$.

11. $y = \left(4 - x^{2/3}\right)^{3/2}$ from $x = 0$ to $x = 8$.

BONUS: Use calculus and the general slicing method to find a general formula for the volume of a pyramid with a square base.